Azure status history

This page contains Post Incident Reviews (PIRs) of previous service issues, each retained for 5 years. From November 20, 2019, this included PIRs for all issues about which we communicated publicly. From June 1, 2022, this includes PIRs for broad issues as described in our documentation.

Product:

All

Region:

All

Date:

All

June 2020

RCA - Storage - East US

Tracking ID: 9VHK-J80

Summary of impact: Between 11:57 and 14:20 UTC on 11 Jun 2020, a subset of Storage customers in East US may have experienced connection failures when trying to access some of their resources hosted in this region. Services with dependencies on the impacted storage resources, such as Virtual Machines, may also have experienced downstream impact during this time.

Root Cause: Engineers determined that an incident during a planned power maintenance activity at the datacenter caused an impact to a single storage scale unit, which then became unhealthy. The incident caused power to be lost to to a subset of racks comprising 60% of this single storage scale unit.

The maintenance activity itself did not impact the storage scale unit, but it caused the scale unit to have reduced redundant power options at the time of the incident. All racks and network devices have two sources of power for redundancy, but it is standard procedure in some types of maintenance to isolate some resources to a single source of power for a short period. After the isolation had been completed on this scale unit, but before maintenance could begin, a distribution breaker in the redundant power source tripped open unexpectedly and the power was lost to the subset of racks.

Mitigation: The site engineers paused all maintenance work and inspected the electrical distribution system to ensure there were no apparent equipment fault issues. They found the tripped breaker and determined it had failed. Power was restored by closing the other breaker that had previously been opened to commence the isolation for the scale unit, and this restored a single power source to the impacted racks. A new breaker was located and fully-tested before installation. The bad breaker on the redundant power supply was replaced with the new breaker and redundant power was then also restored to the affected racks.

11

Once power was restored to the impacted storage racks, the automated restart process for storage resources began, and restored the scale unit to full operation. The restart process for storage clusters follows a series of structured steps to ensure full integrity of customers' data is preserved, and access to storage resources on this scale unit would have become available over a short period of time. Final mitigation was declared at 14:20, but most customers would have seen recovery prior to this time.

Subsequent testing showed that the breaker had an internal failure on one phase and it has been sent to the manufacturer for full forensic analysis.

Next Steps: We sincerely apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- Forensic investigation into the breaker that failed.
- Review of breaker maintenance and testing requirements
- Improving monitoring and alerting visibility when in maintenance modes.
- Ensuring that electrical maintenance activities associated power switching are monitored closely with automated processes to manage unintended impact.

Provide Feedback: Please help us improve the Azure customer communications experience by taking our survey: https://aka.ms/AzurePIRSurvey

RCA - Azure Resource Manager - Failures creating or deleting resources

Tracking ID: DLZG-7C0

Summary of Impact: Between 07:45 and 16:57 UTC on 04 Jun 2020, a subset of customers across all Public Azure regions may have experienced deployment failures when attempting to create or delete certain service based resources via Azure Resource Manager (ARM) deployment and management service due to an underlying networking issue. While the related networking resources for the impacted services were actually being created or deleted during this time, ARM was not notified of the deployment status and hence was failing the service creation or deletion. This issue may have impacted some GET or READ action on the resources. Less than .01% of users would have experienced this issue.

This issue was initially detected an hour after the impact start time and was identified and escalated by an underlying service experiencing end user impact. Once detected, multiple engineering teams were engaged to investigate the cause of the issue to understand what needed to be fixed. By 11:00 UTC, the appropriate networking team was engaged and began investigating. The underlying cause was identified by 13:00 UTC. We identified the appropriate fix and rolled it out to a single region to validate success. We confirmed success of the roll out and began deploying to other regions in 3 batches. At the end of each batch we validated the success of the fix. By 16:57 UTC, the fix was rolled out to all regions and mitigation was confirmed.

Root Cause: A recent ARM deployment contained a configuration file that stores the URL endpoint that ARM connects to for operation status query calls. The configuration file had an incorrect endpoint for networking resources. Due to this wrong setting, the ARM status query for networking service management operations failed, which customers saw as failures when attempting to create or delete networking resources. The faulty configuration file was not caught prior to production because the update that caused the network resource failures was applied after testing was performed on a then healthy configuration file. When picking up the latest configuration file for deployment, the faulty file was assessed for production and not testing. The faulty configuration file was then manually rolled out without testing being performed with the newest configuration, breaking change.

Mitigation: We corrected the incorrect URL endpoint within the configuration file and safely re-deployed to mitigate the issue.

Next Steps: We apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- Networking service will onboard to SDP (Safe Deployment Practice) endpoint configuration rollout process immediately, to ensure enough testing is done and enough time occurs between deployment batches to catch any misconfigurations or changes prior to deployment.
- Networking service will immediately plug-in testing and monitoring holes to make sure we immediately identify an issue like this on the networking end as failures were only seen on the ARM end.
- Networking service will work with ARM team to streamline configuration rollout process, to guard against errors that may occur with the current manual deployment process.

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May 2020

RCA - SQL and Open-Source Database Service Management Issues - East US

Tracking ID: 8K76-LZ8

Summary of Impact: Between 13:30 and 16:30 UTC on 19 May 2020, a subset of customers in East US may have intermittently experienced timeouts and latency issues when processing service management operations such as create, update, delete, for Azure SQL Database, Azure SQL Database Managed Instance, Azure Database for MariaDB, Azure Database for MySQL, Azure Database for

PostgreSQL, and Azure Synapse Analytics services. Some customers may have also encountered issues or experienced latency when loading database management tools or expanding database resources in SQL Server Management Studio (SSMS). Retries may have been successful.

19

Root Cause: Engineers determined that a bug caused high resource utilization in the internal cluster service that is responsible for receiving and executing service management operations in the East US region. The bug was encountered in all the service instances of the region leading to failures and timeouts for management operations.

Mitigation: Engineers mitigated the high resource consumption by identifying and addressing the trigger condition for the bug. System returned to normal after pending operations were drained from the queue.

Next Steps: We apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- Issuing a hotfix for the bug.
- Increasing the resiliency of the SQL service management plane to prevent service disruptions due to high resource utilization.
- Improving the early detection mechanisms for high resource utilization issues.

Provide Feedback: Please help us improve the Azure customer communications experience by taking our survey: https://aka.ms/AzurePIRSurvey

RCA - Multiple Services - Central India

Tracking ID: SLN3-HD0

Summary of Impact: Between 12:41 UTC on 18 May 2020 and 08:30 UTC on 19 May 2020, customers may have experienced difficulties connecting to resources hosted in Central India. A number of Storage and Compute scale units had gone offline, impacting Virtual Machines and other Azure services with dependencies on these.

Root Cause: At 11:25 UTC, a power issue with the regional utility power provider caused a Central India datacenter to transfer to generator power. This transition to generators worked as designed for all infrastructure systems except for the package air handling units (PAHU) in two of the datacenter's colocation rooms (colos). As a result, air temperatures for these two colos rose above operational thresholds. Alerts were triggered and on-site teams began efforts to recover the PAHUs. Initial efforts to recover the PAHU's were unsuccessful, and at 13:22 UTC, engineers began shutting down compute, network and storage resources to protect datacenter equipment from thermal damage.

Mitigation: Engineers undertook various workstreams to bring back connectivity. First, technicians determined why the PAHU's all shut down. By 16:31 UTC engineers implemented a workaround for a failed component and sequentially restarted each unit to bring

18

temperatures back into the safe operating range. In parallel, engineers prepared for equipment to come back online. By 16:48 UTC, temperatures were monitored, deemed stable, and the process of service recovery began. Some network devices and servers needed to be manually powered on due to the manner in which they were powered off. Network was recovered first, with Storage clusters recovering automatically once powered on with the network available. After Storage and Networking were restored, dependent Compute hardware recovery began and came back online by 19:59 UTC allowing Virtual Machines and other dependent Azure services to start recovery and complete by 08:30 UTC on 19 May. Currently, all colos are running on utility power and all PAHU's are running under automatic control.

Next Steps: We sincerely apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- Forensic investigation into the PAHU components to determine the cause of the shutdown.
- Once the forensic investigation is complete, the elimination of the fault from the system through replacement, engineering out, or logical changes.
- If the fault was component-based and systemic, ensure the replacement in other datacenters if present.
- Develop a comprehensive load shedding plan that will keep high priority services up longer to allow for a graceful shutdown.

Provide Feedback: Please help us improve the Azure customer communications experience by taking our survey: https://aka.ms/AzurePIRSurvey

Azure Connectivity - United Kingdom - Mitigated

Tracking ID: Z_V3-FC8

11

Summary of Impact: Between 13:30 and 13:50 UTC on 11 May 2020, a limited subset of customers in the United Kingdom may have experienced connectivity issues to Azure sites and services.

Preliminary Root Cause: Engineers have confirmed the issue was caused by a power event at a regional internet service provider network exchange.

Mitigation: Engineers have confirmed that power was restored, restoring access to resources from the impacted networks and mitigating the issue.

Next Steps: We apologize for any inconvenience this may have caused. Stay informed about Azure service issues by creating custom service health alerts: https://aka.ms/ash-videos for video tutorials and https://aka.ms/ash-alerts for how-to documentation.

April 2020

29

Issues accessing resources in the Azure Portal - Mitigated

Tracking ID: PMN6-7D8

Summary of Impact: Between 18:41 UTC on 29 Apr 2020 and 11:00 UTC on 30 Apr 2020, a subset of customers may have experienced issues accessing resources in their Azure Portal and received an "Access Denied" error message. Programmatic methods, such as PowerShell or Azure CLI, could be used to list resources on their subscriptions and access their individual resources via resource ID.

Preliminary Root Cause: Engineers determined that a recent deployment task introduced a software bug which impacted the synchronization of Role-Based Access Control (RBAC) information for resource management services (such as Azure Resource Manager) which are leveraged by the Azure Portal to display resources. This manifested in Azure Portal call failure and resources not displayed as expected.

Mitigation: Engineers first developed and deployed a hotfix to the impacted regions, forcing a refresh of all RBAC-related data for customers in the region and subsequently scaling-out to handle the increased load.

Next Steps: We apologize for the impact to affected customers. Engineers will continue to investigate to establish the full root cause and prevent future occurrences. Stay informed about Azure service issues by creating custom service health alerts: https://aka.ms/ash-videos for video tutorials and https://aka.ms/ash-alerts for how-to documentation.

22

RCA - Managed Database services - China North

Tracking ID: ZK36-9P8

Summary of Impact: Between 17:37 and 21:54 CST on 22 Apr 2020, a subset of customers may have seen issues affecting service management operations for Azure SQL Database, Azure SQL Database Managed Instance, Azure Database for MariaDB, Azure Database for MySQL, Azure Database for PostgreSQL and Azure Synapse Analytics services in China North. Service management operations including create, rename, update and delete may have been impacted. Connectivity to database resources was not impacted.

Root cause: Engineers determined that a manual maintenance operation impacted instances of an internal cluster data service that is responsible for receiving and executing service management operations. The primary instance of the data service became unhealthy preventing some of the service management operations from completing.

Mitigation: Engineers paused the maintenance operation and initiated failover of the data service's primary to a healthy instance, thus mitigating the issue. Engineers monitored for an extended period post-mitigation to ensure there were no further occurrences. Engineers also worked to complete the maintenance operations offline and restore all instances to operational rotation.

Next steps: We sincerely apologize for the impact to the affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and to our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- Engineers will continue to review the specific maintenance operation to understand the cause of the impact, and will ensure the operation is fully automated and covered by pre-deployment test cases.

RCA - Managed Database services - UK South

Tracking ID: TS66-1C0

Summary of Impact: Between 09:37 and 13:54 UTC on 22 Apr 2020, a subset of customers may have seen issues affecting service management operations for Azure SQL Database, Azure SQL Database Managed Instance, Azure Database for MariaDB, Azure Database for MySQL, Azure Database for PostgreSQL, Azure Database for MySQL, and Azure Synapse Analytics services in UK South. Service management operations including create, rename, update and delete may have been impacted. Connectivity to database resources was not impacted.

Root cause: Engineers determined that a manual maintenance operation impacted instances of an internal cluster data service that is responsible for receiving and executing service management operations. The primary instance of the data service became unhealthy preventing some of the service management operations from completing.

22

Mitigation: Engineers paused the maintenance operation and initiated failover of the data service's primary to a healthy instance, thus mitigating the issue. Engineers monitored for an extended period post-mitigation to ensure there were no further occurrences. Engineers also worked to complete the maintenance operations offline and restore all instances to operational rotation.

Next steps: We sincerely apologize for the impact to the affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and to our processes to help ensure such incidents do not occur in the future.

- Engineers will continue to review the specific maintenance operation to understand the cause of the impact, and will ensure the operation is fully automated and covered by pre-deployment test cases.

Provide Feedback: Please help us improve the Azure customer communications experience by taking our survey: https://aka.ms/TS66-1C0

March 2020

RCA - Service Availability Issue in West Central US

Tracking ID: SMR4-D90

Summary of Impact: Between approximately 20:30 UTC on 15 Mar 2020 and 00:30 UTC on 16 Mar 2020, a subset of customers in West Central US may have experienced issues connecting to resources hosted in this region.

Root Cause: At approximately 20:30 UTC on 15 Mar 2020, an ice storm in the region resulted in a loss of power to both primary and secondary utility power feeds in West Central US. This caused the datacenter to transition to generator power. Although largely successful, a subset of generators failed and UPS units supporting the downstream racks carried the load until their batteries drained. At that point, the affected racks lost power entirely. Any customer resources hosted on the affected racks became unavailable at the time.

15

Mitigation: Site engineers manually intervened to restore power to the affected infrastructure. Concurrently, impacted Azure services implemented their disaster recovery plans to work around the power failure. Subsequently, engineering worked to recover customer resources as affected infrastructure started to become available again.

Next Steps: We sincerely apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this included (but was not limited to):

- Reviewing the sequence of operations and associated control systems to remediate identified single point failures
- Validating electrical hardware components to ensure proper functionality

Provide Feedback: Please help us improve the Azure customer communications experience by taking our survey: https://aka.ms/SMR4-D90.

RCA - Service Management/Authentication Errors - Azure China

Tracking ID: SND4-L80

Summary of Impact: Between 21:03 CST (UTC+8) on 05 Mar 2020 and 16:03 CST on 06 Mar 2020, a subset of customers in the Azure China regions may have encountered failures when performing service management operations on resources hosted in these regions. Customers may also have experienced authentication failures when attempting to access the Azure portal or other Azure resources in the Azure China regions.

Root Cause: When clients connect to an Azure service, they validate the Transport Layer Security (TLS) certificate of that Azure service. This validation requires access to an Online Certificate Status Protocol (OCSP) service and Certificate Revocation List (CRL).

Azure in China uses an external Certificate Authority (CA). This CA hosts the OCSP and CRL endpoints on remote locations. Those endpoints were not reachable from clients in China during the incident. This issue was in a telecom provider and it impacted both Azure and other customers of the CA. It caused clients to fail certificate validation, which in turn caused failure in connecting to Azure services.

These OCSP and CRL endpoints are mirrored in multiple locations via a Content Distribution Network (CDN).

Mitigation: To mitigate, engineers updated DNS records to re-route clients to alternate OCSP and CRL endpoints that were reachable. The troubleshooting took time as multiple companies were involved in the network path. Some Azure services were able to mitigate sooner by deploying the latest CRL to their servers out-of-band.

Next Steps: We sincerely apologize for the impact to affected customers. We are continuously taking steps to improve the Microsoft Azure Platform and our processes to help ensure such incidents do not occur in the future. In this case, this includes (but is not limited to):

- Work with our telecom partner in China to address the root cause of why the primary CRL and OCSP endpoints were not reachable.
- Improve the resiliency of our certificate infrastructure, including hosting the CRL and OCSP endpoints closer to Azure servers.

Provide feedback: Please help us improve the Azure customer communications experience by taking our survey https://aka.ms/SND4-L80

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English (UK)

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